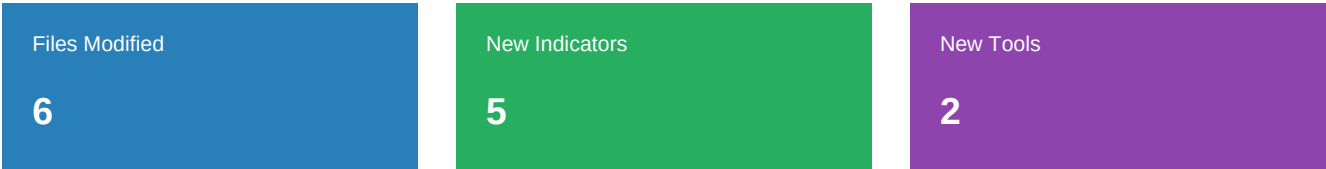


Crypto Daytrading Arena

Profitability Improvements - Change Report

March 2026



This report summarizes seven high-impact improvements to the multi-agent crypto daytrading arena. The changes address the biggest profitability leaks: overtrading (fee bleed), limited technical analysis, insufficient historical data, and agents' lack of self-awareness about their own performance. Together, these changes give agents better data, smarter guardrails, and more order types to execute on.

1. New Technical Indicators

`indicators.py`

Added five new indicator functions and wired them into the summary that agents receive with every market data update. Volume-based indicators are the single best confirmation signal for crypto moves - they help agents distinguish real breakouts from fakeouts.

VWAP (Volume-Weighted Average Price)

Calculates the average price weighted by volume across all candles in the window. When price is above VWAP, buying pressure dominates; below VWAP indicates selling pressure. Computed for both 1-min and 5-min timeframes.

OBV & OBV Trend (On-Balance Volume)

Cumulative volume indicator: adds volume on up-candles, subtracts on down-candles. The trend function compares recent vs. older OBV to determine if volume is RISING, FALLING, or FLAT - confirming or contradicting price direction.

RSI Divergence Detection

Detects bullish divergence (price makes lower low while RSI makes higher low) and bearish divergence (price makes higher high while RSI makes lower high). One of the most reliable reversal signals in crypto trading.

Consecutive Red Candle Counter

Counts consecutive red (close < open) candles from the most recent. Triggers a "falling knife" warning at 3+ candles, directly feeding into the anti-pattern rules.

2. Expanded Candle Data Windows

`coinbase_consumer.py`

The previous candle windows were too narrow for reliable indicator computation. RSI(14) on 20 candles is barely statistically meaningful. The expanded windows give indicators much more data to work with, producing more reliable signals.

Timeframe	Before	After	Gain
1-minute	20 candles (20 min)	60 candles (1 hour)	3x
5-minute	14 candles (with gap)	24 candles (2 hours)	1.7x
15-minute	6 candles (with gap)	24 candles (6 hours)	4x

Gaps between timeframes have also been eliminated. Previously, the 5-min window started at 90 minutes ago and ended at 20 minutes ago, leaving a blind spot. All windows now extend to the present (`end_minutes_ago = 0`).

3. Trade Frequency Guardrails

`trading_tools.py`

Hard-coded guardrails in the trading engine prevent agents from overtrading. At 0.2% round-trip cost, an agent making 50 trades/day bleeds ~10% in fees alone. These guardrails are enforced at the engine level - agents cannot override them.

Maximum 10 trades per hour

A rolling 1-hour window tracks trade timestamps per agent. Trades beyond the limit are rejected with a clear error message. This prevents fee bleed from hyperactive agents.

2-minute minimum hold time

Sell orders are rejected if the position was bought less than 120 seconds ago. The only exception is a stop-loss override: if the position is down more than 1.5% from entry, the sell is allowed regardless of hold time.

4. Performance Tracking in Portfolio

`trading_tools.py`

Agents previously had no awareness of their own track record. Now the portfolio view includes comprehensive performance statistics, enabling agents to self-regulate their risk based on actual results.

New metrics tracked per agent:

- Win rate - percentage of profitable round-trip trades (wins / total sells)
- Average P&L per trade - mean realized profit/loss across all closed positions
- Total realized P&L - cumulative net profit from all completed trades
- Max drawdown - largest peak-to-trough decline as a percentage
- Consecutive losses - current streak of losing trades (triggers risk reduction)
- Trades used this hour - shows proximity to the 10/hour rate limit

The anti-pattern rules reference these stats directly: "If your win rate is below 40%, become more selective" and "After 2 consecutive losses, reduce position size by 50%."

5. Limit Order Support

`trading_tools.py` `tools_and_dashboard.py` `deploy_router_node.py`

Previously, agents could only execute market orders - always buying at the ask and selling at the bid, guaranteeing spread loss on every trade. Limit orders let agents set entries at support levels and exits at resistance levels.

New tools available to agents:

- `place_limit_order(product_id, quantity, action, limit_price)` - queues an order that fills automatically when price reaches the target
- `cancel_limit_order(order_id)` - removes a pending order from the queue

Limit orders are checked against live prices on every price update via the Kafka subscriber. Buy limits fill when the ask drops to the limit price; sell limits fill when the bid rises to the limit price. Pending orders are shown in the portfolio view.

6. Anti-Pattern Rules in Strategy Prompts

```
deploy_router_node.py
```

A shared anti-pattern block is now appended to all four strategy system prompts (default, momentum, contrarian, swing). These rules tell agents what NOT to do - which is often more valuable than entry signals.

Rules enforced via prompt:

- No buying falling knives - wait for a green candle after 3+ consecutive red candles
- No selling dips within uptrends - don't sell if 5-min SMA(5) hasn't been broken
- No trading when spread > 0.5% - transaction costs would eat any edge
- Reduce position size by 50% after 2 consecutive losses
- No chasing moves > 1% in 5 minutes - wait for pullback
- Self-regulate based on win rate - become more selective below 40%
- Prefer limit orders over market orders for entries at key technical levels

7. Event-Driven Agent Invocation

```
coinbase_kafka_connector.py
```

Agents previously received data only on a fixed interval (typically 60 seconds). Fast moves - breakouts and crashes - could be completely missed between intervals.

Now, when any product's price moves more than 0.5% since the last published update, the connector immediately invokes agents with fresh data. This ensures agents can react to significant moves in near real-time while still avoiding noise from small price fluctuations.

Files Modified

File	Changes
indicators.py	New VWAP, OBV, RSI divergence, red candle counter; wired into summary
coinbase_consumer.py	Expanded TIMEFRAMES: 1-min (60 candles), 5-min (24), 15-min (24)
trading_tools.py	Guardrails, performance tracking, limit order engine + tools
deploy_router_node.py	Anti-pattern rules, limit order mentions, new indicator references
tools_and_dashboard.py	Registered place_limit_order & cancel_limit_order; order fill on price update
coinbase_kafka_connector.py	Event-driven invocation on 0.5% price moves

Impact Summary

These seven changes work together as a system. Better indicators give agents more signal. Expanded data windows make those signals statistically reliable. Guardrails prevent the most common profitability killers (overtrading, panic selling). Performance tracking creates a feedback loop for self-regulation. Limit orders reduce spread costs. Anti-pattern rules encode hard-won trading wisdom. And event-driven invocation ensures agents don't miss the moves that matter most.